HDOC DAY – 48

ALGORITHM:

1. Start.
2. Read the size n of the array.
3. Declare a variable-length integer array arr[n].
4. Read n elements into the array.
5. Pass the address of arr and n to pivot_index().
6. In pivot_index():
    - Initialize totalSum = 0.
    - Find the sum of all elements and store it in totalSum.
    - Initialize leftSum = 0.
    - For every index i from 0 to n-1:
        - Calculate rightSum = totalSum - leftSum - arr[i].
        - If leftSum == rightSum, display i as the pivot index and return.
        - Otherwise, add arr[i] to leftSum.
7. If the loop finishes without finding a pivot, display -1.
8. Stop.

C PROGRAM:

```
#include <stdio.h>


void pivot_index(int *arr, int n)
{
   int totalSum = 0;
   int leftSum = 0;
   int rightSum;

   // Find total sum of array
```

```
    for (int i = 0; i < n; i++)
    {
        totalSum += *(arr + i);
    }

    // Check each index
    for (int i = 0; i < n; i++)
    {
        // Right sum = total - left sum - current element
        rightSum = totalSum - leftSum - *(arr + i);

        if (leftSum == rightSum)
        {
            printf("Pivot index: %d\n", i);
            return;    // leftmost pivot found
        }

        // Add current element to left sum
        leftSum += *(arr + i);
    }

    printf("-1\n");
}

int main()
{
    int n;
```

```
    printf("Enter the size of array: ");
    scanf("%d", &n);

    int arr[n];    // Variable Length Array

    printf("Enter the array elements:\n");

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    // Passing address of array
    pivot_index(arr, n);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Input Array	Expected Output
1	1 7 3 6 5 6	Pivot index: 3
2	1 2 3	-1
3	2 1 -1	Pivot index: 0
4	1 -1 0	Pivot index: 0
5	5	Pivot index: 0